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PCN Frontier Review

Interface between hypothalamic-pituitary-adrenal axis and brain-derived neurotrophic factor in depression

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Although the pathophysiology of depressive disorder remains elusive, two hypothetical frameworks seem to be promising: the involvement of hypothalamic pituitary-adrenal (HPA) axis abnormalities and brain-derived neurotrophic factor (BDNF) in the pathogenesis and in the mechanism of action of antidepressant treatments. In this review, we focused on research based on these two frameworks in relation to depression and related conditions and tried to formulate an integrated theory of the disorder. Hormonal challenge tests, such as the dexamethasone/corticotropin-releasing hormone test, have revealed elevated HPA activity (hypercortisolism) in at least a portion of patients with depression, although growing evidence has suggested that abnormally low HPA axis (hypocortisolism) has also been implicated in a variety of stress-related conditions. Several lines of evidence from postmortem studies, animal studies, blood levels, and genetic studies have suggested that BDNF is involved in the pathogenesis of depression and in the mechanism of action of biological treatments for depression. Considerable evidence has suggested that stress reduces the expression of BDNF and that antidepressant treatments increase it. Moreover, the glucocorticoid receptor interacts with the specific receptor of BDNF, TrkB, and excessive glucocorticoid interferes with BDNF signaling. Altered BDNF function is involved in the structural changes and possibly impaired neurogenesis in the brain of depressed patients. Based on these findings, an integrated schema of the pathological and recovery processes of depression is illustrated.

Key words: hypothalamic-pituitary-adrenal axis (HPA axis), neurotrophic factor, plasticity, stress, TrkB.

MOOD DISORDERS ARE common diseases with a lifetime prevalence of 2–20% for major depression and 0.3–1.5% for bipolar disorder worldwide.[1] In Japan, the 1-year prevalence of major depression was estimated to be as high as 2.9%.[2] Mood disorders also comprise a leading cause of suicide.[3] Depressive disorders are one of the top-ranked diseases in terms of the adjusted life years burden of disease (DALY).[4] Although pathogenesis and pathophysiology of mood disorders remain elusive, two hypothetical frameworks seem to be promising: the involvement of hypothalamic pituitary-adrenal (HPA) axis abnormalities and brain-derived neurotrophic factor (BDNF) in the pathogenesis and in

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the mechanism of action of antidepressants. The former plays an important role in stress response and the latter in structural and functional changes in the brain responsible for mood disorders. In this review, we focused on research based on these two frameworks in relation to mood disorders and tried to formulate an integrated theory of the disorders.

HPA AXIS AND DEPRESSION

Since the seminal work of Selye,[5] a wide variety of stress has been associated with an activation of the hypothalamic-pituitary-adrenal (HPA) axis. Stress-related psychiatric disorders, including major depression, have also been reported to be associated with alteration in HPA axis function. Indeed, abnormality in HPA axis function is one of the most extensively studied biological markers for depression.[6]

HPA axis

Stressors of all sorts, both physical and psychological, activate the HPA axis by increasing the production and release of corticotropin-releasing hormone (CRH) and arginine vasopressin (AVP) from the paraventricular nucleus of the hypothalamus. Via the portal vein system, CRH, in concert with AVP, stimulates the pituitary to produce adrenocorticotropic hormone (ACTH), which enters the bloodstream and activates the adrenal glands to release glucocorticoids (cortisol in primates, including humans, and corticosterone in rodents). Glucocorticoids, in turn, exert inhibitory feedback effects mainly at the hypothalamus and pituitary glands to inhibit the synthesis and secretion of CRH and ACTH, respectively. Hippocampus also confers an inhibitory effect on HPA axis.

Various measures to monitor HPA axis

In addition to baseline studies, several challenge paradigms have been developed to characterize HPA axis function. These can be divided into psychosocial challenge tests, such as the Trier Social Stress Test (TSST),[7] and pharmacological ones, such as the dexamethasone (DEX) suppression test (DST)[8] and the DEX/CRH test.[9,10]

The TSST, a standardized psychosocial stress test, has been extensively used in the field of psychoneuroendocrinology. This test consists of a 3-min preparation phase followed by a 5-min free speech phase (job interview) and a 5-min mental arithmetic task in front of an audience. In this test the subjects' self-esteem is threatened by a committee that pretends to evaluate the subjects' performance, which leads to the feelings of uncontrollability. A meta-analysis on acute laboratory stressors found the TSST to be a reliable tool to elicit robust physiological stress responses.[11]

To quantify the dysregulation of the HPA axis, the DST, mostly using 1 mg of DEX, has been extensively studied since Carroll *et al.* standardized it as a biological marker for the diagnosis of melancholia.[12] In a series of DST studies, cortisol levels as measured by the DST were shown to be increased in depressed patients.[13] However, it has subsequently become clear that its sensitivity to differentiate depressed patients from healthy controls is not very high,[14,15] and elevated cortisol levels were also observed in non-clinical populations under various stressful conditions.[16,17] The DST has thus failed to fulfill the initial promise as a diagnostic tool for depression. On the other hand, more recent studies that employed DST with lower doses of DEX (e.g. 0.5 mg) have reliably identified enhanced negative feedback in several psychiatric disorders, including post-traumatic stress disorder, chronic fatigue syndrome (CFS) and fibromyalgia.[18–20]

The DEX/CRH test was developed in an attempt to enhance the sensitivity of the DST.[9,10] It is an integrated provocative test for HPA axis function that combines DEX pretreatment with CRH administration on the following day; thus, it is essentially a DST followed by CRH challenge. In the standard protocol of the DEX/CRH test, a relatively high dose (i.e. 1.5 mg) of DEX is used. The merit of this combined test is that at the moment of CRH infusion the HPA axis is downregulated due to negative feedback induced by DEX. This test has been shown to better discriminate depressive patients from healthy people compared to the original DST.[10,21–23] Using the DEX/CRH test, abnormalities in HPA axis function in several other psychiatric disorders, such as bipolar disorder, panic disorder and personality disorders, have also been reported.[24–26] Furthermore, this test, like DST, has been increasingly used to detect the enhanced suppression (or blunted reactivity of cortisol) in varied conditions.

The prednisolone suppression test (PST) is a newly developed pharmacological challenge test to measure HPA axis function.[27] The investigators proposed a test using 5 mg of prednisolone, which gave

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approximately 30–40% suppression of salivary cortisol in healthy volunteers, as a useful tool to investigate negative feedback inhibition of the HPA axis. The foundations of this test are the similarities between prednisolone and cortisol in terms of their similar half-lives as well as their similar abilities to bind to and activate both the glucocorticoid receptor (GR) and mineralocorticoid receptor (MR), whereas DEX has a much longer half-life and binds solely to GR. Based on these facts, the authors argue that PST is an ideal test to investigate both enhanced and impaired negative feedback of the HPA axis in patients with psychiatric disorders.[27] A prospective study of patients with depression showed that non-response to inpatient treatment was predicted by a more dysfunctional HPA axis as indicated by higher post-prednisolone cortisol levels.[28] In PST for patients with CFS, significantly greater suppression of both salivary cortisol and urinary cortisol metabolites was observed in CFS patients compared to controls.[29]

HPA axis abnormalities in depression

There is mounting evidence for an important role of HPA axis abnormalities in the pathophysiology of mood disorders.[30] Numerous studies have associated melancholic depression with increased HPA axis activity as revealed, for example, by elevated concentrations of CRH in cerebrospinal fluid, increased volumes of adrenal gland and pituitary, and a higher rate of non-suppression to the DST and the DEX/CRH test.[10,13,31–33] Among depressive disorders, previous DST as well as DEX/CRH studies have observed pronounced HPA axis hyperactivity in psychotic depression.[34,35] Moreover, it is suggested that the DST and the DEX/CRH test could be used as a state-dependent biomarker for depression; in DST studies, conversion from the non-suppressor to suppressor is temporally associated with clinical responses to antidepressants[36,37] and hormonal responses to the DEX/CRH test also tend to restore after successful treatment with antidepressants.[38,39] Using the DEX/CRH test, we examined the HPA axis function in hospitalized depressed patients and found that their cortisol responses were significantly greater than those of healthy controls.[21,22] Such abnormal cortisol responses were improved after inpatient treatment, particularly in those patients who underwent electroconvulsive therapy (ECT) in addition to pharmacotherapy as compared to pharmacotherapy alone (Fig. 1).[22] These results suggested that the DEX/CRH test could serve as a state-dependent marker to monitor HPA axis abnormalities in major depressive episodes. Of particular note, several studies have reported that both in- and outpatients remitted from a major depressive episode who still exhibit exaggerated cortisol responses to the DEX/CRH test are at greater risk for relapse than their counterparts with ameliorated cortisol responses.[40–43]

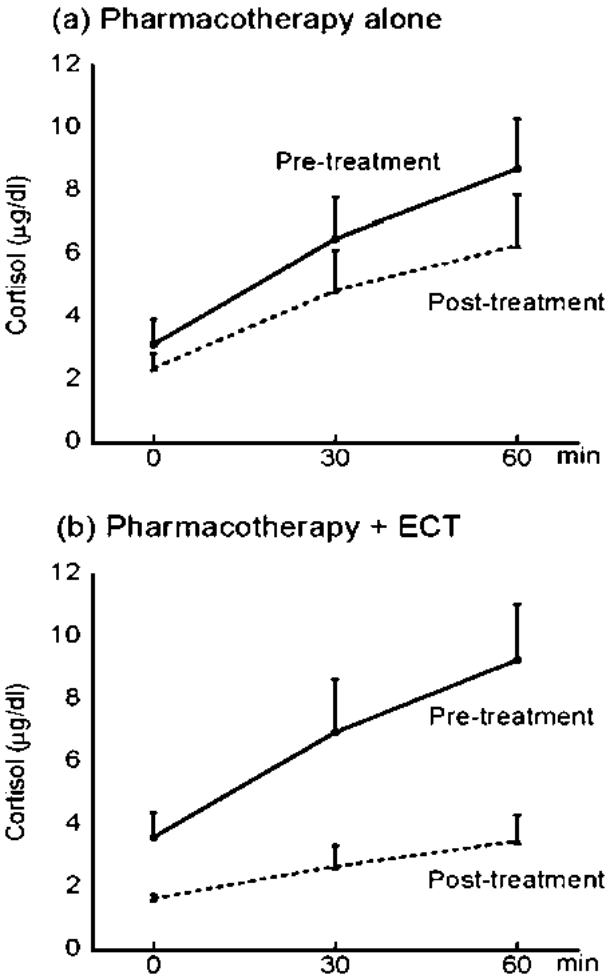

Figure 1. Time-course curves of cortisol responses to the dexamethasone/corticotropin-releasing hormone (CRH) test before and after treatment in (a) the pharmacotherapy group ($n = 23$) and (b) the electroconvulsive therapy (ECT) group ($n = 12$). The *X*-axis represents time after CRH infusion. Error bars represent standard errors of the mean. Adapted from Kunugi *et al.*[22]

In contrast, a number of recent studies using the DEX/CRH test have reported that depressed patients

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as a whole show similar,[44–47] or even attenuated, cortisol responses as compared to healthy controls.[48–50] In these studies, patients were one of the following: outpatients,[45–47] chronically depressed patients,[44] depressive patients with psychiatric comorbidity,[50] or long-term sick-leave patients.[48,49] These inconsistent findings on the DEX/CRH test in depressed patients are likely to result from the heterogeneity of depression as well as from a general role of stress in HPA axis function. Indeed, atypical depression, in contrast to melancholic depression, is suggested to relate to hypocortisolism rather than hypercortisolism.[51] Other stress-related psychiatric conditions characterized by hypocortisolism include post-traumatic stress disorder, CFS, and fibromyalgia.[52,53]

BDNF AND DEPRESSION

BDNF function

BDNF belongs to the neurotrophin family, including nerve growth factor (NGF), neurotrophin-3 (NT-3), and NT-4/5, which bind to high-affinity Trk receptors as well as to a common low-affinity p75 receptor. These neurotrophins play important roles in growth, differentiation, maintenance, death/survival, and plasticity of neurons. Interestingly, Trk receptors contain a tyrosine kinase domain that exerts trophic effects, whereas p75 belongs to the tumor necrosis factor family and has a death domain that plays a role in apoptosis as well. Among the neurotrophins, BDNF and its specific receptor TrkB are highly expressed in the adult brain and they are essential in survival of neurons and neurotransmission. In immature neurons, BDNF is involved in growth, differentiation, maturation, and survival, while it plays an important role in synaptic plasticity, augmentation of neurotransmission, and regulation of receptor sensitivity in mature neurons (reviewed in Numakawa *et al.*[54]). BDNF is translated as a precursor protein (proBDNF) and then proteolytical1y c1eaved (processed) to generate a small mature protein (mBDNF). The p75 neurotrophin receptor binds to proneurotrophin with high affinity.[55] BDNF has therefore the yang and yin in the action on neurons depending on processing of proBDNF and differential affinity of proBDNF and mBDNF for TrkB and p75 receptors.[56] Binding of BDNF to TrkB leads to activation of the receptor through phosphorylation, which induces several intracellular signaling pathways, i.e. mitogen-activated protein kinase (MAPK), phospholipase C-gamma (PLCγ), and phosphatidylinositol 3-kinase (PI3K) pathways.[57]

Evidence for BDNF in depression

Studies on postmortem brains of depressed patients and blood levels of BDNF in depressed patients have suggested the important role of BDNF in depression (reviewed in Duman and Monteggia[58]). In postmortem studies of suicide victims with depression, BDNF expression has quite consistently been reported to be reduced in the hippocampus.[59–61] Such reduction was also observed in the prefrontal cortex.[60] Importantly, BDNF expression was unchanged or even increased in the hippocampus of suicide victims with antidepressants,[60,62] which suggests that antidepressants increase the level of BDNF. As regards blood BDNF levels, drug naïve patients with depression often showed decreased BDNF, while they were increased in patients treated with antidepressants (e.g. Shimizu *et al.*[63]). Recent meta-analyses confirmed such findings.[64,65] Further, a significant correlation was found between changes in BDNF level after antidepressant medications and changes in depression scores.[65] However, the possible use of blood BDNF level as a biomarker for depression needs further studies because BDNF exists abundantly in platelets and it is still unclear how the blood level of BDNF reflects that in the brain.

Genetic evidence for BDNF in mood disorders

Growing evidence from molecular genetic studies has also suggested that genetic variations in the BDNF gene confer susceptibility to mood disorders. There are at least two functional polymorphisms in the BDNF gene that have been extensively studied in relation to neuropsychiatric diseases. The most well-studied polymorphism is the single nucleotide polymorphism (SNP) of A758G (rs6265) in the coding region resulting in an amino acid change of Val66Met in the proBDNF protein.[66] Functional characterization of this polymorphism revealed that the Met66 allele was found to be associated with abnormal hippocampal activation and impaired episodic memory in humans.[67] Neurons transfected with Met66-BDNF-GFP showed lower depolarization-induced secretion and the Met66-BDNF-GFP failed to localize to secretary granules or synapses in neurons.

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Transgenic mice that were homozygous for the Met66 allele exhibited increased anxiety-related behaviors that were not normalized by the antidepressant, fluoxetine, suggesting that this variant of the BDNF gene may play a role in genetic predispositions to anxiety and depressive disorders.[68] The same research group subsequently reported that the 66Met homozygous mice showed altered adult olfactory bulb neurogenesis with altered spontaneous olfactory discrimination and impaired extinction of conditioned aversive memory.[69,70] Because of these functional effects of the Val66Met polymorphism it was expected that this polymorphism might be associated with susceptibility to various neuropsychiatric diseases. Indeed, bipolar disorder was initially reported to be associated with this polymosphism.[71,72] However, these studies reported that the Val66 allele, which has been shown to be associated with better BDNF functions than the Met allele, was the risk allele for bipolar disorder. Subsequent studies, including ours, could not replicate the association between the Val66 allele and the risk of bipolar disorder.[73,74] As for major depression, a number of association studies have been conducted, yielding inconsistent results. A recent meta-analysis on 14 studies (2812 cases and 10 843 controls) revealed that the Met66 allele (odds ratio [OR] 1.27, 95% confidence interval [CI]: 1.10–1.47) and homozygosity for the Met66 allele (OR: 1.67, 95%CI: 1.19–2.36) gives a risk of major depressive disorder in men but not in women.[75] Interestingly, such a sexually dimorphic effect of the polymorphism was also observed and the Met66 allele was associated with the risk of Alzheimer's disease in female subjects, but not in male subjects.[76] Pertinent to this, BDNF conditional knockout mice, in which the BDNF gene is deleted selectively in the forebrain, demonstrated sexually dimorphic effects in the opposite direction; male conditional knockouts exhibited normal depression-related behaviors, whereas female conditional knockouts displayed a striking increase in depression-like behavior.[77] Indeed, estrogen plays an important role in the expression of BDNF. Estrogen receptors co-localize with BDNF-synthesizing neurons in the forebrain and estrogen induces BDNF expression through the estrogen response element.[78,79]

Another polymorphism of functional significance is the 'BDNF-linked complex polymorphic region (BDNF-LCPR)' located 1kb upstream (putative promoter region) of the coding exon. This polymorphic site was initially reported as a simple dinucleotide repeat (GT repeat).[80] However, we subsequently characterized this polymorphism because previous studies reported a significant association between this repeat polymorphism and bipolar disorder.[71,81] Surprisingly, we found that this polymorphic site has a very complex structure, but not the simple dinucleotide repeat; it contains three different dinucleotide repeats in succession, yielding a total of 23 novel allelic variants.[82] Among the four common alleles, the 'A1 allele' was found to be associated with reduced transcriptional activity and associated with a risk of bipolar disorder, suggesting that this polymorphism confers susceptibility to bipolar disorder by reducing the transcriptional activity of the BDNF gene.[82]

As regards receptors of BDNF, we identified a non-synonymous SNP (Ser205Leu) within the p75 gene; the minor allele (L205) was significantly decreased in the patients than in the controls ($P < 0.05$, OR 0.54, 95%CI 0.31-0.94), suggesting that this allele may have a protective effect against the development of major depression.[83] Furthermore, this association was more strongly observed in patients with a history of attempted suicide than those without such a history. Therefore, the Ser205Leu polymorphism of the p75 gene might be involved in the pathogenesis of depressive disorder and suicidal behavior.

All these findings suggest that genetic variations of BDNF and its receptor p75 play a role in giving susceptibility to mood disorders.

BDNF in the mechanism of action of antidepressants

A number of studies examined the effect of antidepressants on the expression of BDNF. Although some studies have found negative results, the majority of previous studies support the early findings of Nibuya *et al.*, who reported that chronic, not but acute, administration of antidepressants increases BDNF expression in the hippocampus.[84] Several classes of antidepressants, including tricyclics, selective serotonin reuptake inhibitors (SSRI), serotonin noradrenalin reuptake inhibitors (SNRI) and monoamine oxidase inhibitors (MAOI), have been found to increase BDNF in the hippocampus.[58] ECT and other treatments (e.g. repetitive transcranial magnetic stimulation [rTMS]) have also been demonstrated to increase BDNF.[84,85] In relation to diet and nutrition, foods containing omega-3 fatty acid (e.g. fish oil), which has been shown to be effective

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in the treatment of major depression,[86] increase BDNF in the hippocampus.[87] Thus, BDNF might be involved in the final common pathway of the various antidepressant treatment strategies. The possible mechanism of increased expression of BDNF by antidepressants is such that chronic antidepressant administration increases the expression of cAMP response element-binding protein (CREB), and CREB, a transcriptional factor, then upregulates its target genes, such as BDNF and TrkB.[88] Furthermore, histone acetylation has also been implicated in the link between antidepressants and increased BDNF expression.[89]

Impacts of stress-induced excessive glucocorticoid on BDNF expression and function

There are at least two lines linking altered HPA axis and BDNF dysfunction. First, stress-induced hyperactivity of the HPA axis and resultant increase in glucocorticoid level reduce the BDNF expression. Second, GR, through which glucocorticoid exerts its effects, directly influences the function of the specific receptor of BDNF, TrkB. Many studies demonstrated reduced expression of BDNF in the hippocampus of animals with various kinds of acute and chronic stress (e.g. restraint, footshock, social isolation, social defeat, swim stress, etc.) and early environmental stress (e.g. maternal deprivation) (summarized by Duman and Monteggia[58]). As mentioned above, stress activates the HPA axis and increases the glucocorticoid level, which in turn decreases BDNF expression in the hippocampus.[90,91] Recent studies suggest that stresses, such as immobilization and social defeat, reduce the expression of BDNF via the mechanism of histone remodeling.[89,92]

Although many studies have shown that stress and glucocorticoid regulates expression of BDNF, there is little information on whether excessive glucocorticoid impacts BDNF function. We then examined the effect of glucocorticoid on BDNF-regulated synaptic function in cultured neurons. In immature hippocampal neurons, exposure to glucocorticoid (DEX) inhibited the BDNF-dependent dendrite outgrowth and synaptic formation (Fig. 2).[93] As a result, BDNF-induced synaptic proteins, such as NR2A, NR2B, GluR1, and synapsin I, were suppressed by DEX, and the inhibitory action of DEX influenced neuronal function even after the neurons had matured.[93] Furthermore, our subsequent study elucidated that GR directly interacts with TrkB and promotes BDNF-triggered PLC-γ signaling for glutamate release via glutamate transporter,[94] which raises the possibility that excessive glucocorticoid might decrease the TrkB-bound GR, which leads to decreased BDNF signaling (Fig. 3). As the observed functional effects of glucocorticoid and GR were not accounted for by altered expression of BDNF, the findings show that stress-induced excess in glucocorticoid impacts not only expression level but also function of the BDNF pathway.

BDNF and structural brain changes in depression

Although there are inconsistencies across studies, a smaller hippocampal volume in depressed patients is thought to be related to the pathophysiology of the disease (reviewed in Eker and Gonul[95]). Many studies in major depressed/suicidal subjects have demonstrated altered brain structure, such as reduction in cell number, density, cell body size, neuronal and glial density in frontal cortical or hippocampal brain areas and decrease in parahippocampal cortex cortical/laminar thickness.[96] There is evidence that antidepressants and ECT increase hippocampal volume in patients with depression.[97,98]

Furthermore, there is evidence that TrkB-dependent neurogenesis is involved in the antidepressant effect; mice lacking TrkB in the hippocampal neuron progenitor cells had impaired neurogenesis and proliferation induced by antidepressant treatment. These mice also demonstrated increased anxiety-like behavior and decreased sensitivity to antidepressants.[99,100] X-irradiation of a restricted region of mouse brain containing the hippocampus prevented the neurogenic and behavioral effects of two classes of antidepressants. These findings suggest that the behavioral effects of chronic antidepressants may be mediated by the stimulation of neurogenesis in the hippocampus.[101]

Taken together, BDNF may play a key role in the brains of recovering patients during antidepressant treatment.[102] However, BDNF might not be the sole key molecule; vascular endothelial growth factor (VEGF), for example, plays an important role in neurogenesis and has been found to be decreased in response to stress and increased by antidepressants and ECT.[103]

As mentioned above, hippocampus regulates the negative feedback of the HPA axis. Therefore, hippocampal damage may lead to disruption of this feed-

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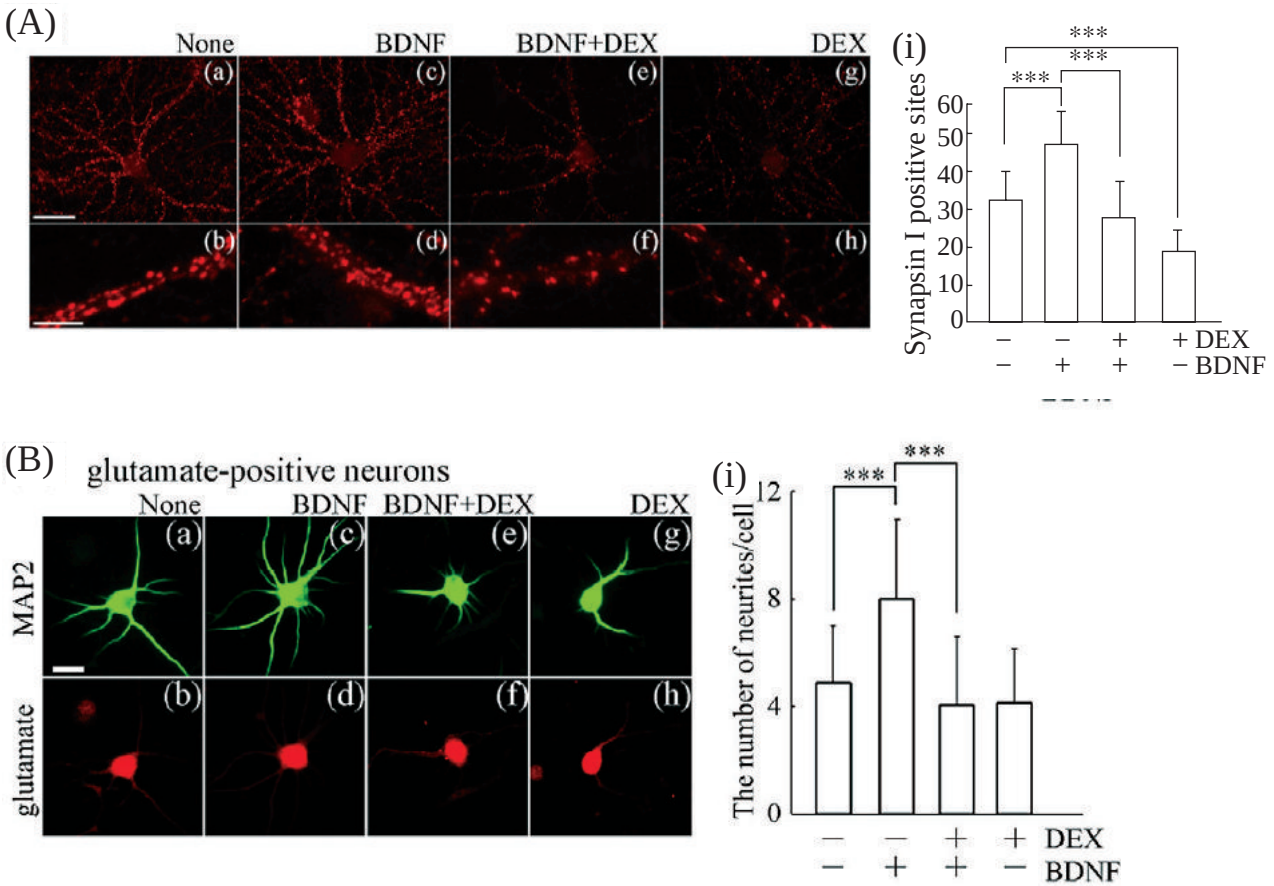

Figure 2. Excessive glucocorticoid (dexamethasone [DEX]) suppresses brain-derived neurotrophic factor (BDNF)-induced neurite outgrowth and synaptic formation in cultured immature hippocampal neurons. (A) The number of presynaptic sites was quantified after immunostaining with anti-synapsin I antibody. Top: Representative images from (a) untreated cultures (none) and (c) cultures pretreated with BDNF, (e) BDNF plus DEX, and (g) DEX. Bar, 20 µm. Bottom: High-magnification images of dendrites from (b) untreated cells (none) and (d) cells pretreated with BDNF, (f) BDNF plus DEX, and (h) DEX. Bar, 10 µm. (i) Quantification of the number of synapsin I-positive presynaptic sites per dendritic shaft (50 µm). Data represent mean ± SD. (B) Effect of DEX on neurite outgrowth of glutamatergic neurons. (a, c, e, and g) MAP2-positive and (b, d, f and h) glutamate-positive glutamatergic neurons are shown (merged images not shown here). (a,b) Untreated cultures (none). (c, d) BDNF increased the number of neurites compared with control, whereas (e,f) DEX significantly suppressed the BDNF-induced neurite outgrowth. (g,h) DEX had no influence compared with control. Bar: 20 µm. (i) Quantification indicates that the number of glutamatergic neurites was increased by BDNF, and the increase was suppressed by DEX. Data represent mean ± SD, $***P < 0.001$; $**P < 0.01$. Adapted from Kumamaru *et al.*[93] with permission.

back loop and resultant continuous hyperactivation of the HPA axis, which in turn further damages the hippocampus. This vicious cycle might be involved in the pathogenesis of depression.

SCHEMA OF DEPRESSION

Based on the roles of the HPA axis and BDNF described above, a possible schema of the pathogenesis and recovery process of depression could be illustrated as in Figure 4. Chronic stress induces hyperactivity of the HPA axis, and resultant excessive stress hormone (glucocorticoid) brings about reduced expression and impaired function of BDNF, which damages the hippocampus and other brain areas. As the hippocampus regulates the feedback system of the HPA axis, its damage further augments the HPA axis activity, which could form a vicious

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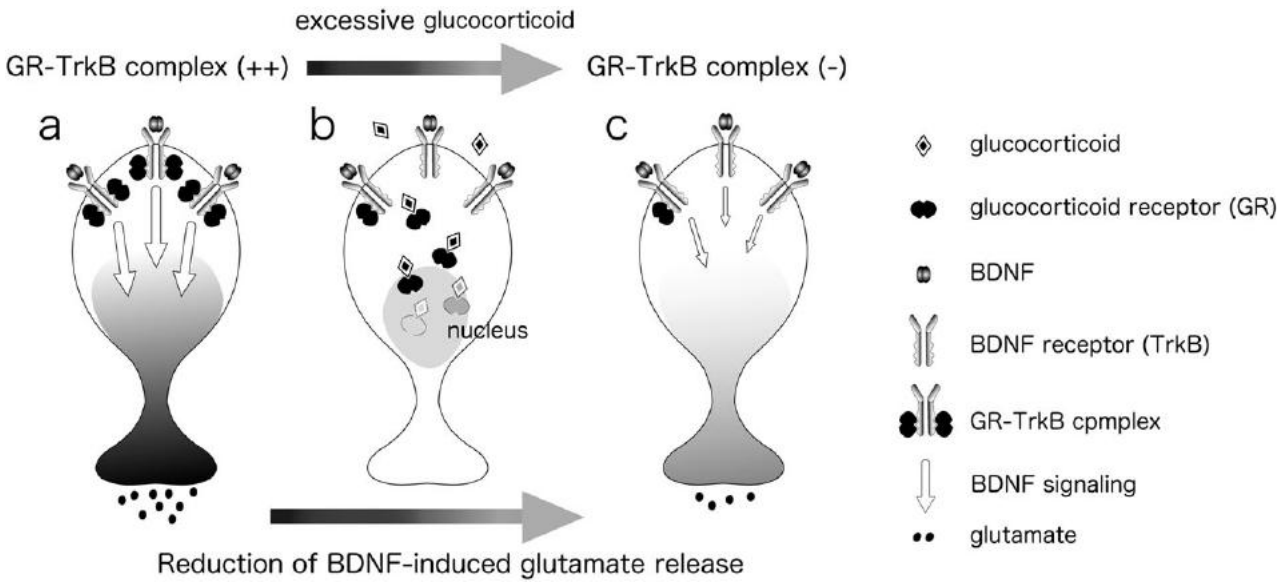

Figure 3. Excessive glucocorticoid might decrease the TrkB-bound glucocorticoid receptor (GR) and decrease brain-derived neurotrophic factor (BDNF) signaling and BDNF-induced glutamate release. (a) When GR-TrkB complex is rich, the BDNF-TrkB signaling for glutamate release is also rich. (b) Exposure to excessive glucocorticoid reduces TrkB-bound GR, which leads to (c) reduced BDNF-TrkB signaling and glutamate release.

cycle. Depressive disorder may develop in the process. In the treatment of depression, relief from stressful situations and biological treatments (e.g. antidepressant medication and ECT) lead to reduction in the hyperactivity of the HPA axis and activation of BDNF and other neurotrophic factors, both of which facilitate each other. Such process will reinstate damaged hippocampus and other brain areas involved in the development of depression.

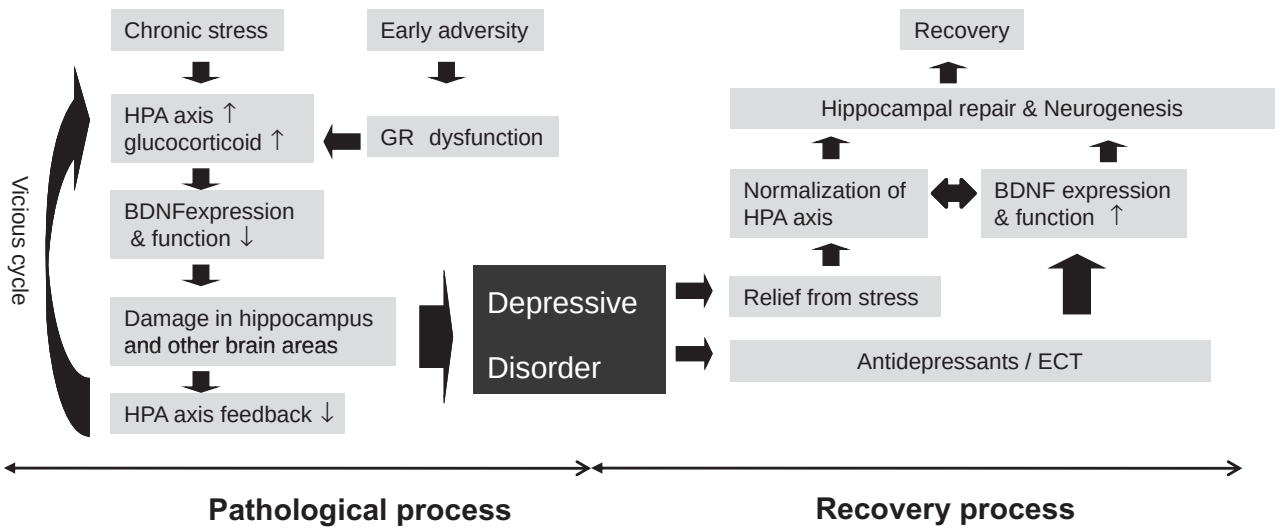

Figure 4. Schematic illustration of pathological and recovery processes of depression focusing on the roles of the hypothalamic pituitary-adrenal (HPA) axis and brain-derived neurotrophic factor (BDNF). ECT, electroconvulsive therapy.

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This schema is a rather simplified model and many other factors are likely to be involved. Furthermore, there are some types of depression that clearly do not accord with this schema, atypical depression, for example. As mentioned above, atypical depression has been suggested to be associated with hypoactivity rather than hyperactivity of the HPA axis,[51] and it is relatively resistant to antidepressant treatment. Moreover, there are certain studies that are not always consistent with the BDNF hypothesis in depression.[104] As depression is a heterogenous condition, further studies on the roles of the HPA axis and BDNF in the illness will lead to a valid classification of the illness based on the key molecules.

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